

20 Practical Tasks on NumPy, Pandas, Matplotlib & Seaborn

1. Create Arrays

Create:

- 1D array from 1 to 20
- 5×5 matrix of zeros
- Identity matrix

2. Random Numbers

Generate:

- 50 random integers between 1–100
- Find mean, max, min

3. Array Operations

Perform:

- Addition
- Subtraction
- Multiplication
- Division between two arrays

4. Reshape Array

Create array from 1–36 and reshape into:

- 6×6
- 3×12

5. NumPy Statistics

Find:

- Mean
- Median
- Standard deviation
- Variance

6. Student DataFrame

Create DataFrame:

- Name
- Marks
- City
- Age

Then:

- Display top 3 rows
- Filter marks greater than 80

7. CSV Analysis

Load a CSV file and:

- Show shape
- Column names
- Missing values
- Data types

8. Sorting Data

Sort employees based on:

- Salary descending
- Age ascending

9. GroupBy Task

Group sales data by category and find:

- Total sales
- Average sales

10. Data Cleaning

Handle:

- Missing values
- Duplicate rows
- Incorrect column names

11. Merge DataFrames

Create two DataFrames and:

- Perform inner join
- Left join
- Right join

12. Pivot Table

Create pivot table for:

- Department vs Salary

13. Line Chart

Plot monthly sales for 12 months.

Add:

- Title
- Labels
- Grid

14. Bar Chart

Compare marks of 5 students.

15. Histogram

Visualize age distribution of 100 students.

16. Pie Chart

Show market share of mobile brands.

17. Multiple Plots

Create:

- One line plot
 - One bar plot
- in same figure.

18. Heatmap

Use correlation matrix and create heatmap.

19. Pairplot

Load iris dataset and create pairplot.

20. Titanic Dataset Analysis

Using Titanic dataset:

- Count males vs females
- Survival comparison
- Age distribution
- Correlation heatmap

Bonus Mini Project: Sales Dashboard

Create a complete dashboard using:

- NumPy for calculations
- Pandas for data handling
- Matplotlib for charts

- Seaborn for advanced visualization

Include:

- Sales trends
- Product analysis
- Heatmaps
- Distribution plots
- KPI metrics